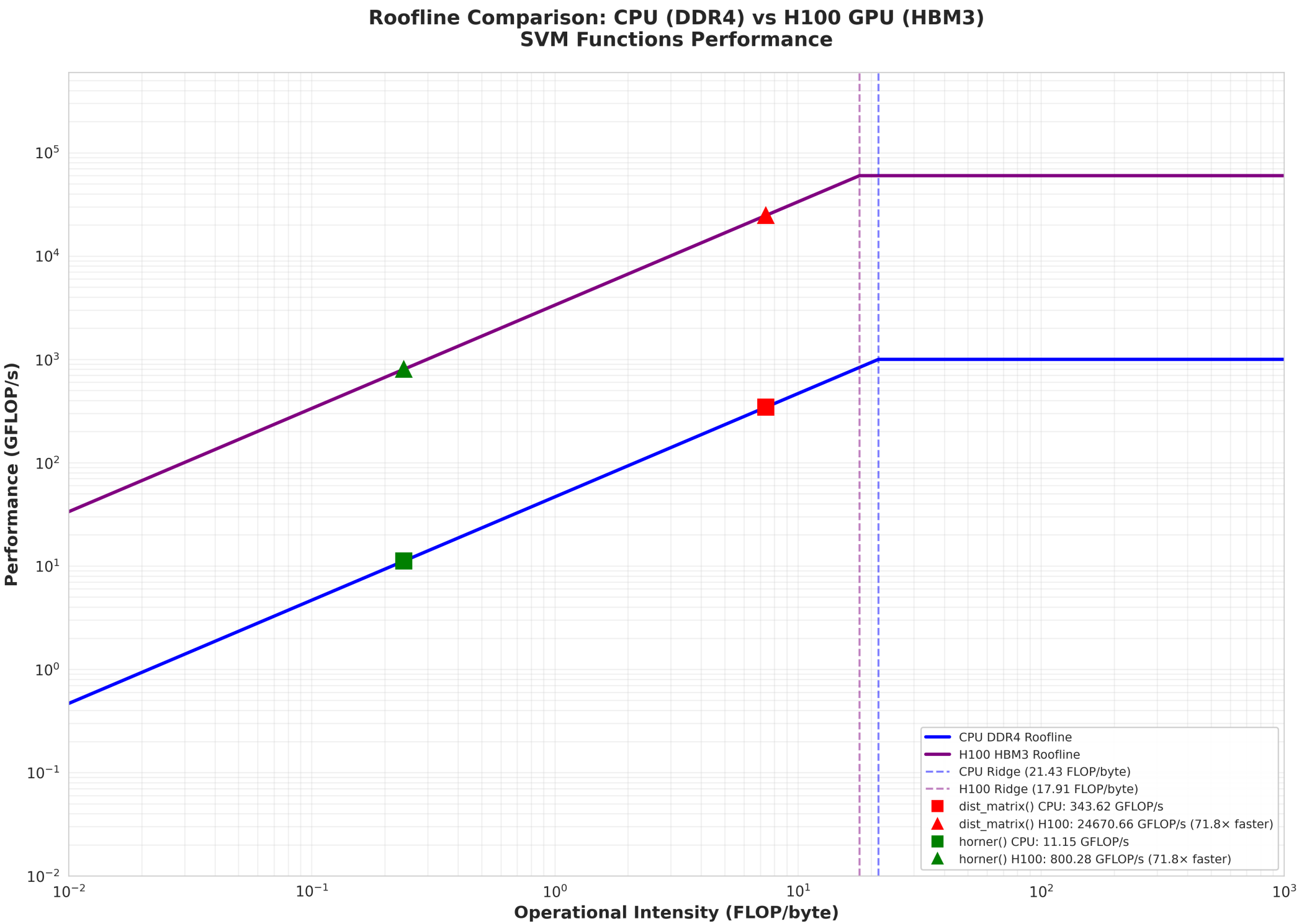
Roofline Comparison: CPU (DDR4) vs H100 GPU (HBM3)
SVM Functions Performance
Performance (GFLOP/s)
Operational Intensity (FLOP/byte)
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
CPU DDR4 Roofline
H100 HBM3 Roofline
CPU Ridge (21.43 FLOP/byte)
H100 Ridge (17.91 FLOP/byte)
dist_matrix() CPU: 343.62 GFLOP/s
dist_matrix() H100: 24670.66 GFLOP/s (71.8× faster)
horner() CPU: 11.15 GFLOP/s
horner() H100: 800.28 GFLOP/s (71.8× faster)